

Code – Lab 4

```
<!DOCTYPE html>

<html>

<head>

<title>Lab 4 - DOM Manipulation</title>

<style>

*{margin:0;padding:0;box-sizing:border-box}

body{font-family:Arial;background:linear-gradient(135deg,#f4e4d7,#faf0e6);padding:2rem}

.h{background:linear-
gradient(135deg,#8B4513,#D2691E);color:white;padding:2rem;border-
radius:12px;margin-bottom:2rem;text-align:center}

.s{background:white;padding:1.5rem;margin-bottom:1.5rem;border-radius:12px;box-
shadow:0 5px 20px rgba(0,0,0,.1)}

.s h2{color:#8B4513;margin-bottom:1rem;border-bottom:3px solid #D2691E;padding-
bottom:.5rem}

button{background:linear-
gradient(135deg,#D2691E,#8B4513);color:white;border:none;padding:.6rem
1.2rem;border-radius:6px;cursor:pointer;font-weight:bold;margin:.3rem}

button:hover{transform:translateY(-2px);box-shadow:0 5px 15px rgba(210,105,30,.3)}

.ob{background:#f9f9f9;border:2px solid #ddd;padding:1rem;border-
radius:8px;margin:1rem 0;min-height:70px}

input,select{padding:.6rem;border:2px solid #ddd;border-radius:6px;margin-right:.3rem}

.gi{display:grid;grid-template-columns:repeat(auto-
fit,minmax(120px,1fr));gap:.8rem;margin:1rem 0}

.gii{background:linear-
gradient(135deg,#D2691E,#8B4513);color:white;padding:1rem;border-radius:8px;text-
align:center;cursor:pointer;transition:.3s;font-size:.9rem}

.gii:hover{transform:scale(1.05)}

.gii.s{transform:scale(1.1);box-shadow:0 0 3px #FFD700}
```

```
.cd{font-size:2.2rem;font-weight:bold;color:#D2691E;text-align:center;padding:1.2rem;background:#fff9e6;border-radius:8px;margin:.8rem 0}
```

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</style>
```

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</head>
```

```
<body>
```

```
<div class="h"><h1> 🍷 DOM Manipulations</h1><p>Create, modify, traverse DOM elements</p></div>
```

```
<div style="max-width:900px;margin:0 auto">
```

```
<div class="s"><h2>1. Create & Remove Elements</h2><input type="text" id="in" placeholder="Item" value="Espresso"><input type="number" id="ip" placeholder="Price" value="3.50" style="width:70px"><button onclick="add()">Add</button><button onclick="clr()" style="background-color:#dc3545">Clear</button><div id="lst" class="ob"><p style="color:#999;margin:0">Items here...</p></div></div>
```

```
<div class="s"><h2>2. Modify Content</h2><button onclick="inc()">+</button><button onclick="dec()">-</button><button onclick="rst()" style="background-color:#6c757d">Reset</button><div class="cd" id="cnt">0</div><div class="ob"><p style="margin:0">Orders: <strong id="oc">0</strong></p></div></div>
```

```
<div class="s"><h2>3. Modify Attributes & Styles</h2><button onclick="col('#8B4513')">Brown</button><button onclick="col('#D2691E')">Orange</button><button onclick="col('#FFD700')">Gold</button><button onclick="col('#2196F3')">Blue</button><div id="cb" style="height:80px;background:linear-gradient(135deg,#8B4513,#D2691E);color:white;display:flex;align-items:center;justify-content:center;border-radius:8px;font-weight:bold;font-size:.9rem;margin:1rem 0">Color: #8B4513</div></div>
```

```
<div class="s"><h2>4. Traverse DOM</h2><div class="gi" id="gc"><div class="gii"> 🍷 </div><div class="gii"> 🍰 </div><div class="gii"> 🍹 </div><div class="gii"> 💎 </div><div class="gii"> 🍷 </div><div class="gii"> 🍰 </div><div class="gii"> 🍹 </div></div></div>
```

```
class="ob"><p style="margin:0">Selected: <strong id="sc">0</strong></p><button
onclick="cls()" style="margin-top:.5rem">Clear</button></div></div>
```

```
<div class="s"><h2>5. Form Values</h2><label>Coffee:</label><select id="cs"
onchange="upd()" style="width:100%;margin-
bottom:1rem"><option>Select</option><option value="Espresso">Espresso -
$3.50</option><option value="Cappuccino">Cappuccino - $4.50</option><option
value="Latte">Latte - $4.50</option><option value="Mocha">Mocha -
$5.00</option></select><div><label><input type="checkbox" id="es" onchange="upd()">
Extra Shot (+$0.50)</label><br><label><input type="checkbox" id="dv" onchange="upd()">
Delivery (+$2.00)</label></div><div class="ob" style="margin-top:1rem"><p
style="margin:.3rem 0">Coffee: <strong id="sd">None</strong></p><p
style="margin:.3rem 0">Extras: <strong id="se">None</strong></p><p
style="margin:0">Total: <strong id="st">$0.00</strong></p></div></div>

</div>
```

```
<script>
```

```
let c=0;
```

```
function add(){const
n=document.getElementById('in').value,p=document.getElementById('ip').value;if(!n||!p)ret
urn;const
l=document.getElementById('lst'),i=document.createElement('div');i.style.cssText='backgr
ound:#fff9e6;padding:.7rem;margin:.3rem 0;border-left:4px solid
#D2691E;display:flex;justify-content:space-between;align-
items:center';i.innerHTML=` <div>  <strong>${n}</strong> $$${p}</div><button
style="background-color:#dc3545;padding:.3rem .6rem;font-size:.8rem"
onclick="this.parentElement.remove()">X</button> `;if(l.children[0]?.textContent.includes
('Items'))l.innerHTML="";l.appendChild(i)}
```

```
function clr(){document.getElementById('lst').innerHTML='<p
style="color:#999;margin:0">Items here...</p>'}
```

```
function inc(){c++;updc()}
```

```
function dec(){if(c>0)c--;updc()}
```

```
function rst(){c=0;updc()}
```

```

function
updc(){document.getElementById('cnt').textContent=c;document.getElementById('oc').tex
tContent=c}

function col(cr){const
b=document.getElementById('cb');b.style.backgroundColor=cr;b.textContent='Color: '+cr}

document.querySelectorAll('.gii').forEach(e=>e.addEventListener('click',function(){this.clas
sList.toggle('s');document.getElementById('sc').textContent=document.querySelectorAll('.
gii.s').length}));

function
cls(){document.querySelectorAll('.gii.s').forEach(e=>e.classList.remove('s'));document.get
ElementById('sc').textContent='0'}

function upd(){const
cf=document.getElementById('cs').value||'None',ex=[],t=cf==='None'?0:parseFloat(cf.matc
h(/\$[\d.]+/)[0]);if(document.getElementById('es').checked){ex.push('Shot');t+=.5}if(docum
ent.getElementById('dv').checked){ex.push('Delivery');t+=2}document.getElementById('sd')
.textContent=cf;document.getElementById('se').textContent=ex.length>0?ex.join(',
'):'None';document.getElementById('st').textContent='$'+t.toFixed(2)}

</script>

</body>

</html>

```

Output –

 **DOM Manipulations**
Create, modify, traverse DOM elements

1. Create & Remove Elements

Items here...

2. Modify Content

0

Orders: 0

3. Modify Attributes & Styles

Color: #8B4513

5. Form Values

Coffee:

☐ Extra Shot (+\$0.50)
☐ Delivery (+\$2.00)

Coffee: None
Extras: None
Total: \$0.00